

# Modeling the context dependence of artifact noun interpretation

*Brandon Waldon, Cleo Condoravdi, Beth Levin,  
and Judith Degen*

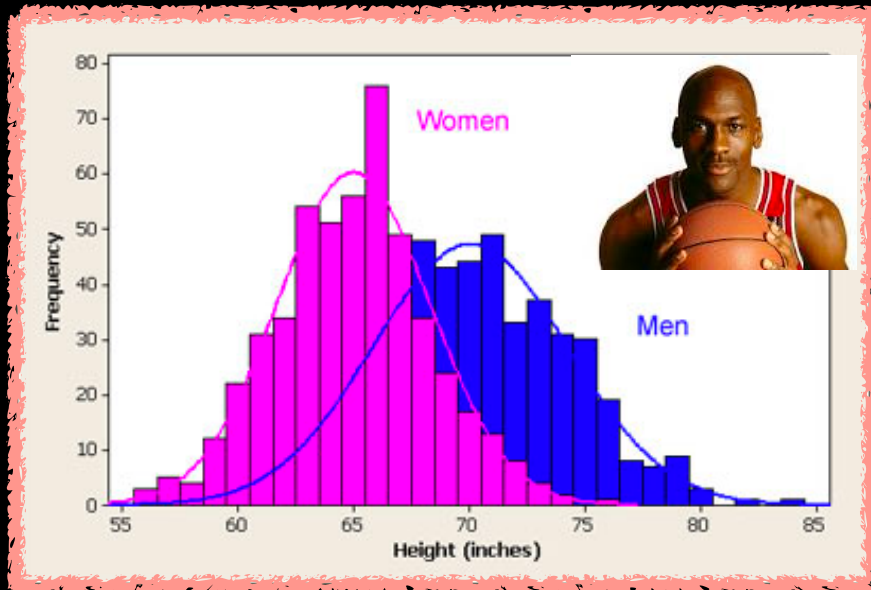
XPRAG 9 —22. September 2022



# Linguistic vagueness

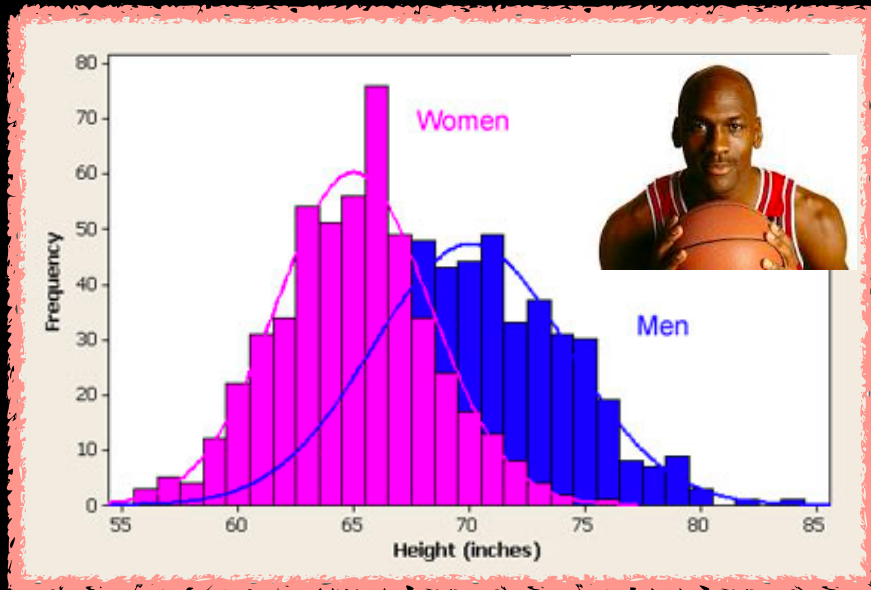
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🙄 *Michael Jordan is tall (for a basketball player)*

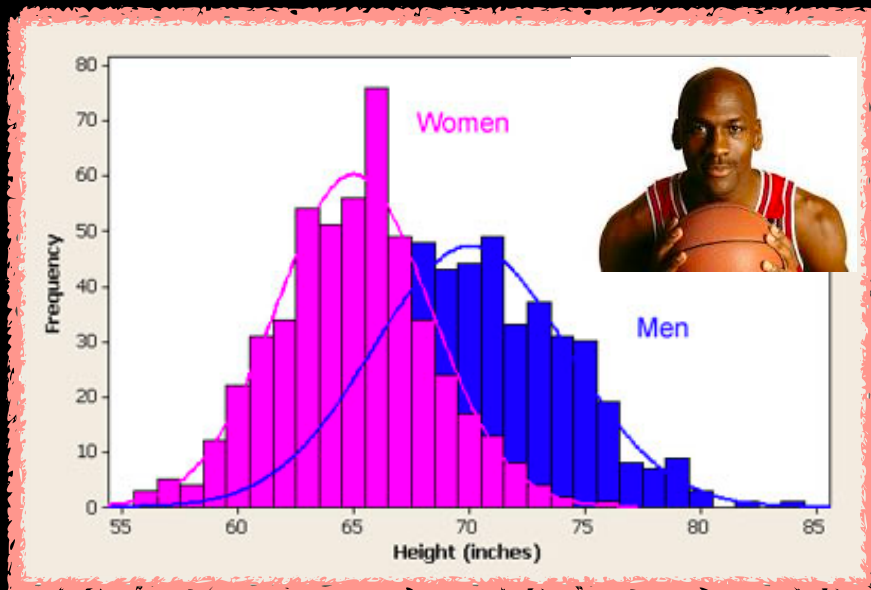


→ Context dependence



# Linguistic vagueness

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→ Context dependence

→ Edge cases

Gradable adjectives (e.g. *tall*):

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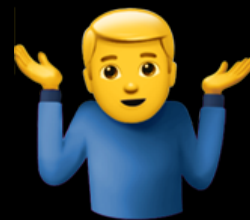


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**Highlighting the function(s) of AN objects cues interpreters re:  
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**Probabilistic model sensitive to  
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(See also Waldon et al., forthcoming in *Proc. Sinn und Bedeutung*)

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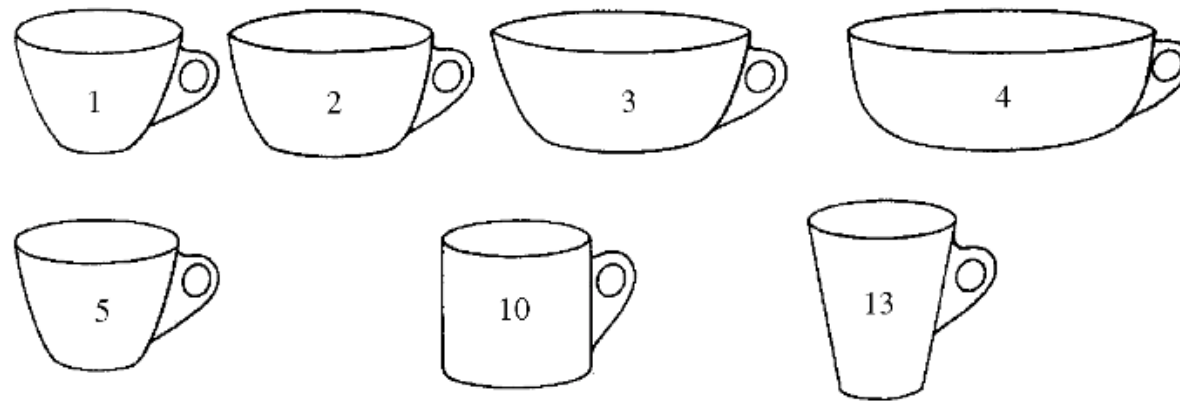
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# Invoking AN function modulates category boundary beliefs

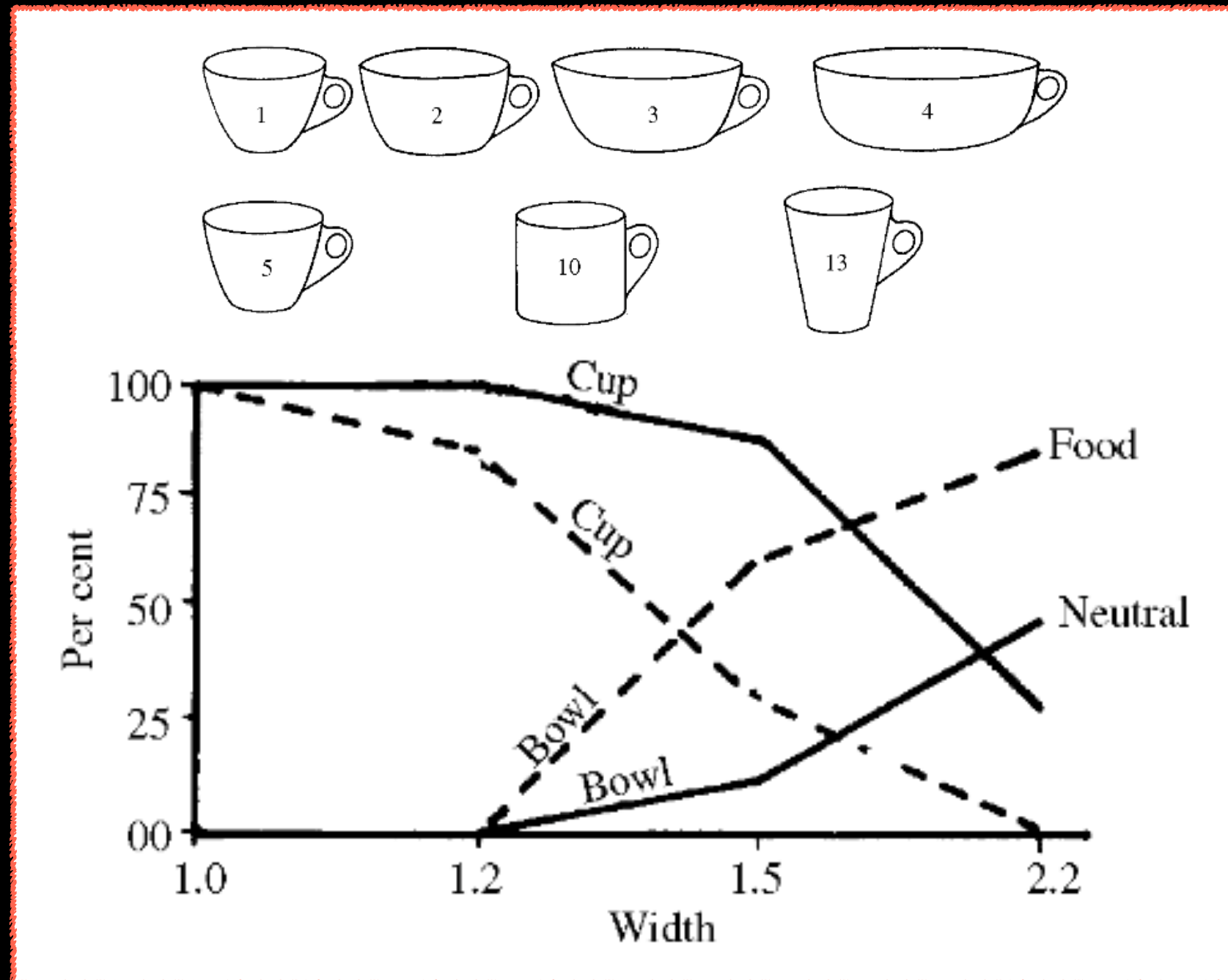
*Cup or bowl?*



Labov (1975)

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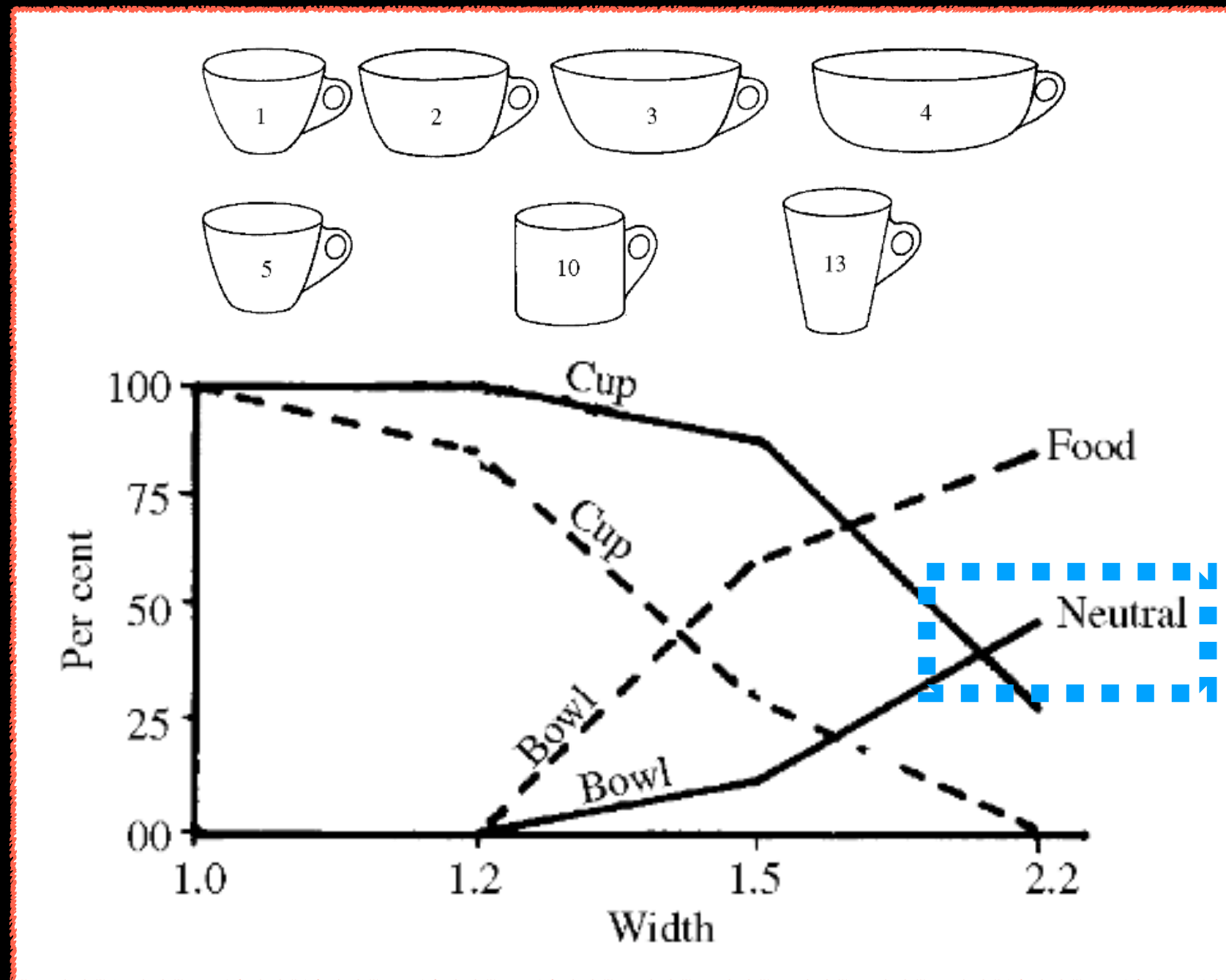
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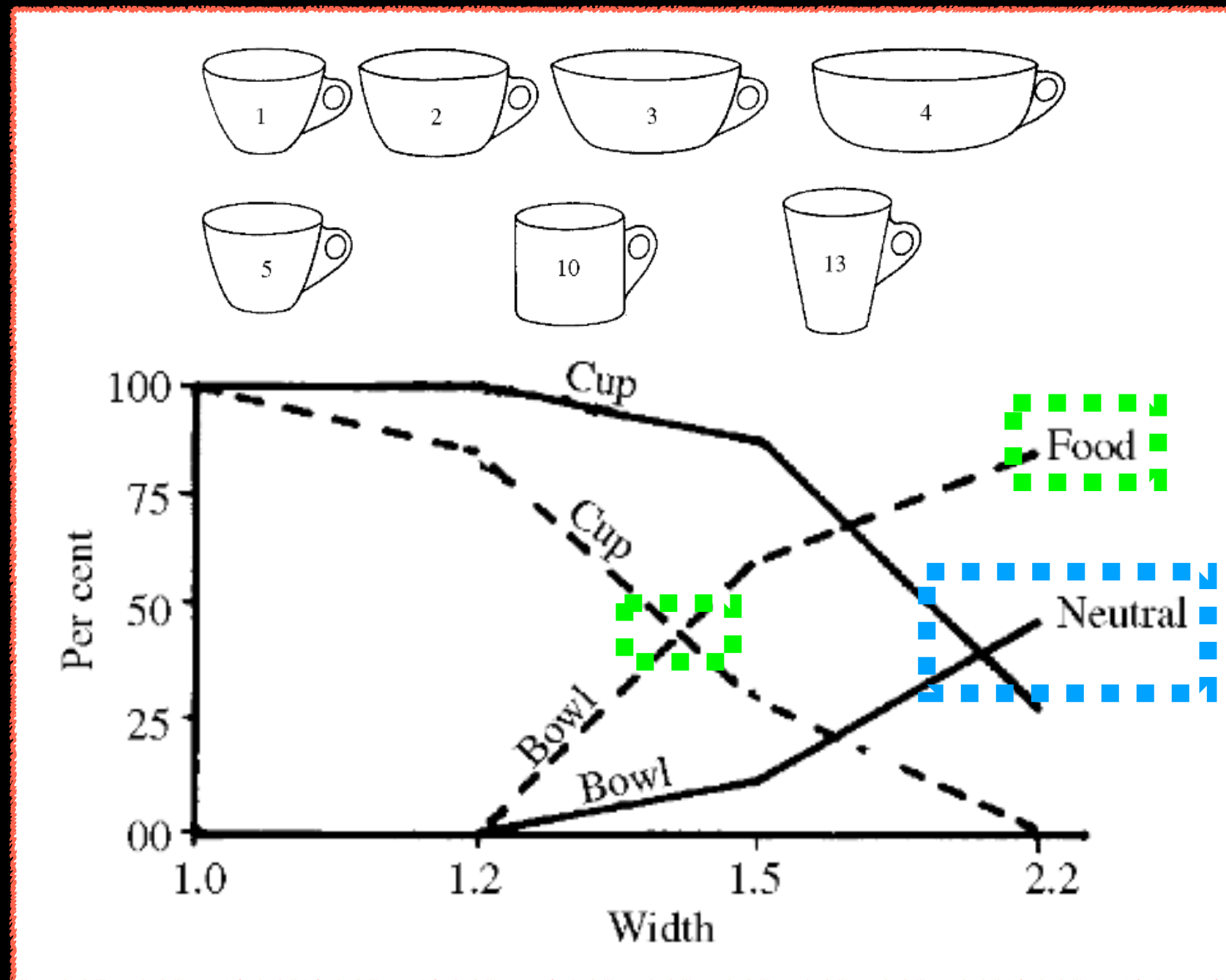
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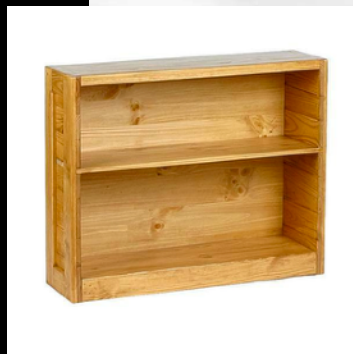


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# Invoking AN function modulates comparison judgements

Who has *more* furniture?

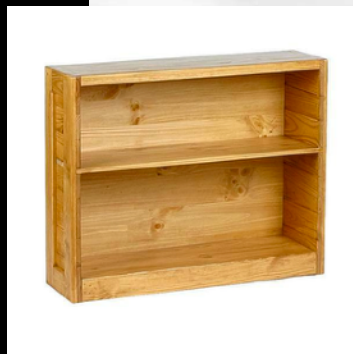


Grimm & Levin (2016)



# Invoking AN function modulates comparison judgements

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**‘Function-neutral’ context**

Grimm & Levin (2016)

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**‘Function-in-effect’ context**

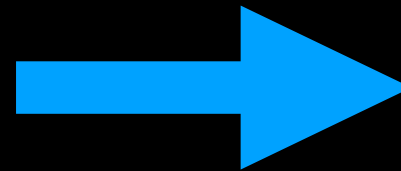


**‘Function-neutral’ context**

Grimm & Levin (2016)

How to systematically invoke  
function-oriented contextual cues  
for a broad class of artifact nouns?

**Policy goal: limit pollution**



**“No *vehicles* allowed...”**



invokes AN function

Policy goal: limit pollution



“No *vehicles* allowed...”





invokes AN function

Policy goal: limit pollution

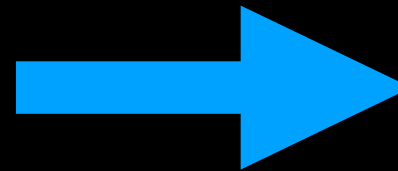


“No *vehicles* allowed...”



invokes AN function

Policy goal: limit pollution



“No *vehicles* allowed...”



Do policy goals influence interpretation of rules targeting AN categories?

(see recent work in *experimental jurisprudence*:  
e.g. Garcia et al. 2014; Struchiner et al. 2020; )

# Experiment

A small town is concerned that there is too much ambient noise in their town square.

Because of this, the local government passed a new rule:

**No vehicles are allowed in the town square.**

Select each item that, if brought into the town square, would violate the rule.





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[Policy goal]

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# Materials

- 12 rules (mix of prohibitions, e.g. *No vehicles allowed in the town square* and requirements, e.g. *Shoes must be worn in the courtyard*)
- 4 between-subjects 'Policy goal' conditions per rule
  - 3 conditions: different policy contexts for each rule (e.g. the 'ambient noise' motivation for *No vehicles*)
  - 1 'goal-neutral' condition

- Every scene contained:

- 'Clear exemplars' of the nominal category



- 'Edge cases'



- 'Clear non-members' of the category



(vehicle)

# ANs tested:

- *electronic devices, face coverings, food, furniture, jewelry, pets, sandwiches, shoes, tents, uniforms, vehicles*

# Hypotheses & predictions

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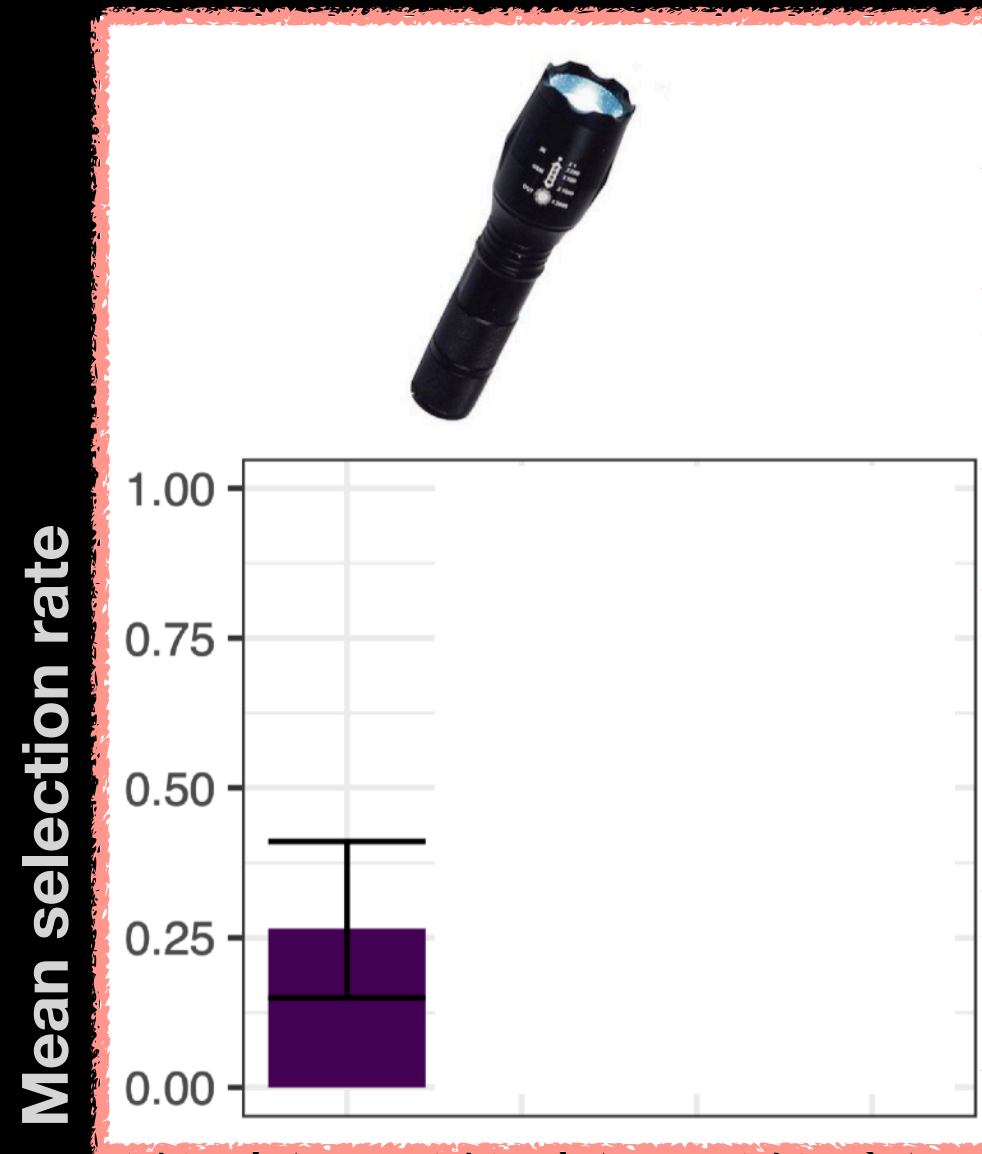
- **Goal-sensitive hypothesis:** interpreters integrate beliefs about policy goals → modulation in selection rate across ‘policy goal’ experiment conditions

# Hypotheses & predictions

- **Goal-sensitive hypothesis:** interpreters integrate beliefs about policy goals → modulation in selection rate across 'policy goal' experiment conditions
- **Goal-insensitive hypothesis:** interpreters don't integrate beliefs about policy goals → no modulation



N = 200  
(188 after  
exclusions)



No electronic devices are allowed in the theater.

Policy  
goal:



None

Prevent pirate recordings

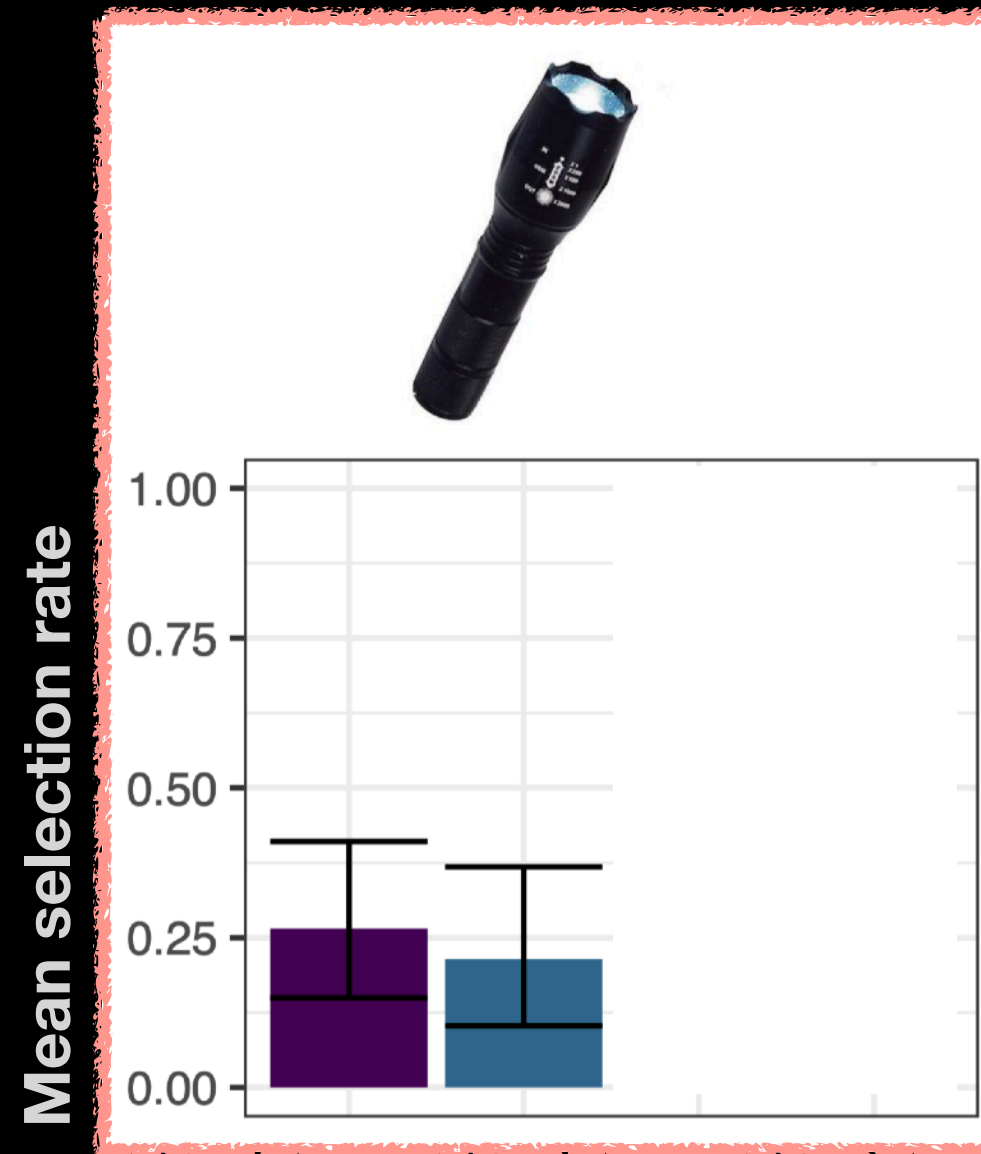


Limit distracting noise

Limit distracting light



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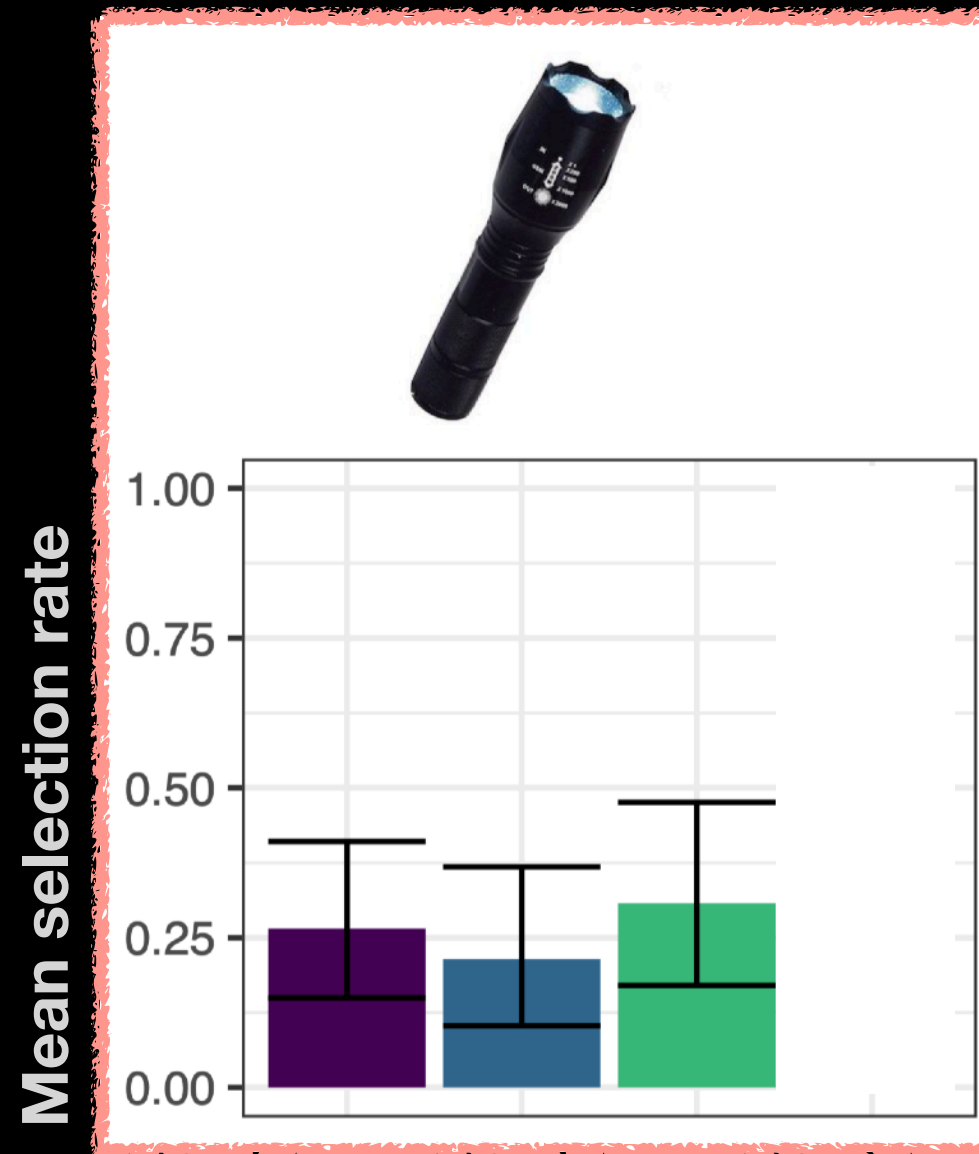
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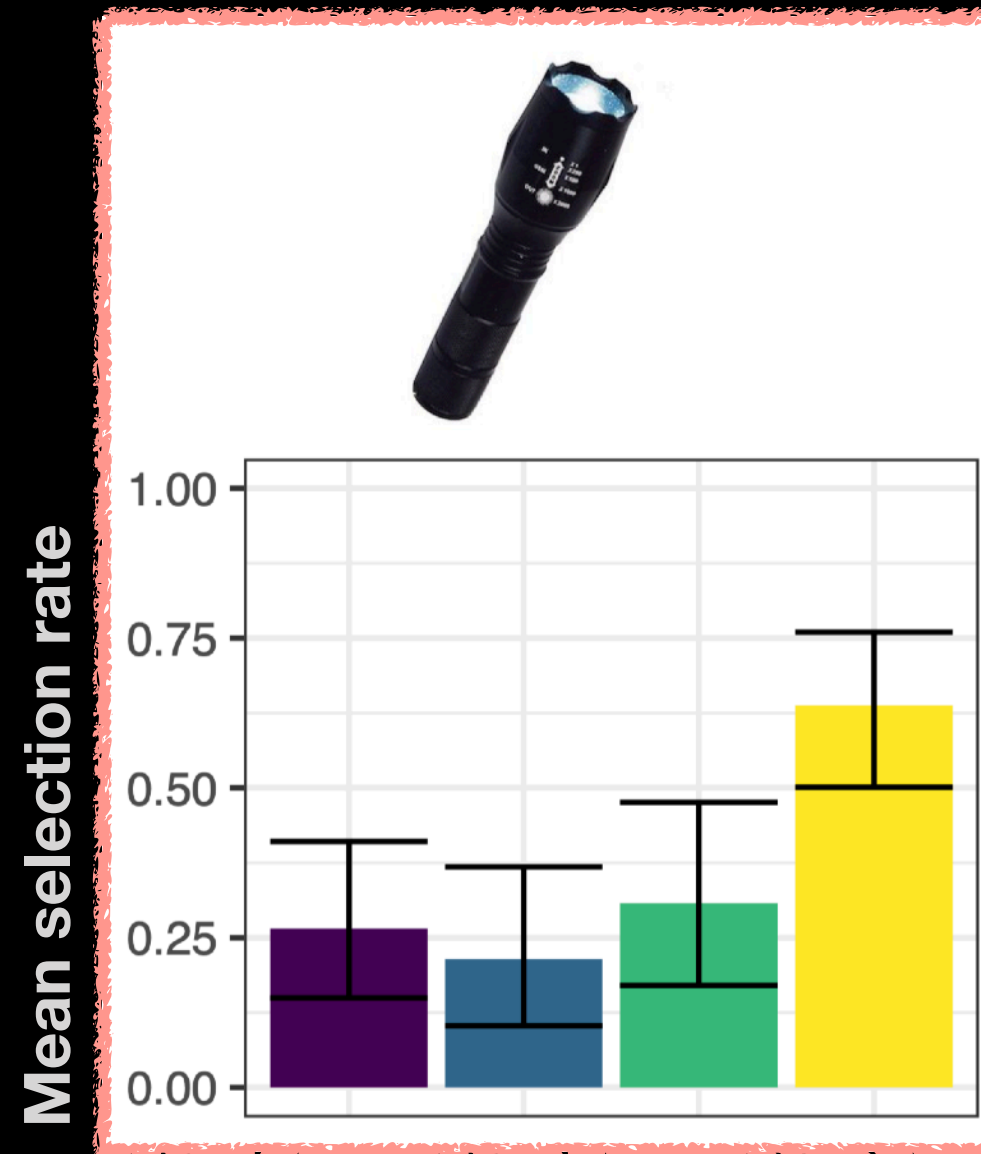
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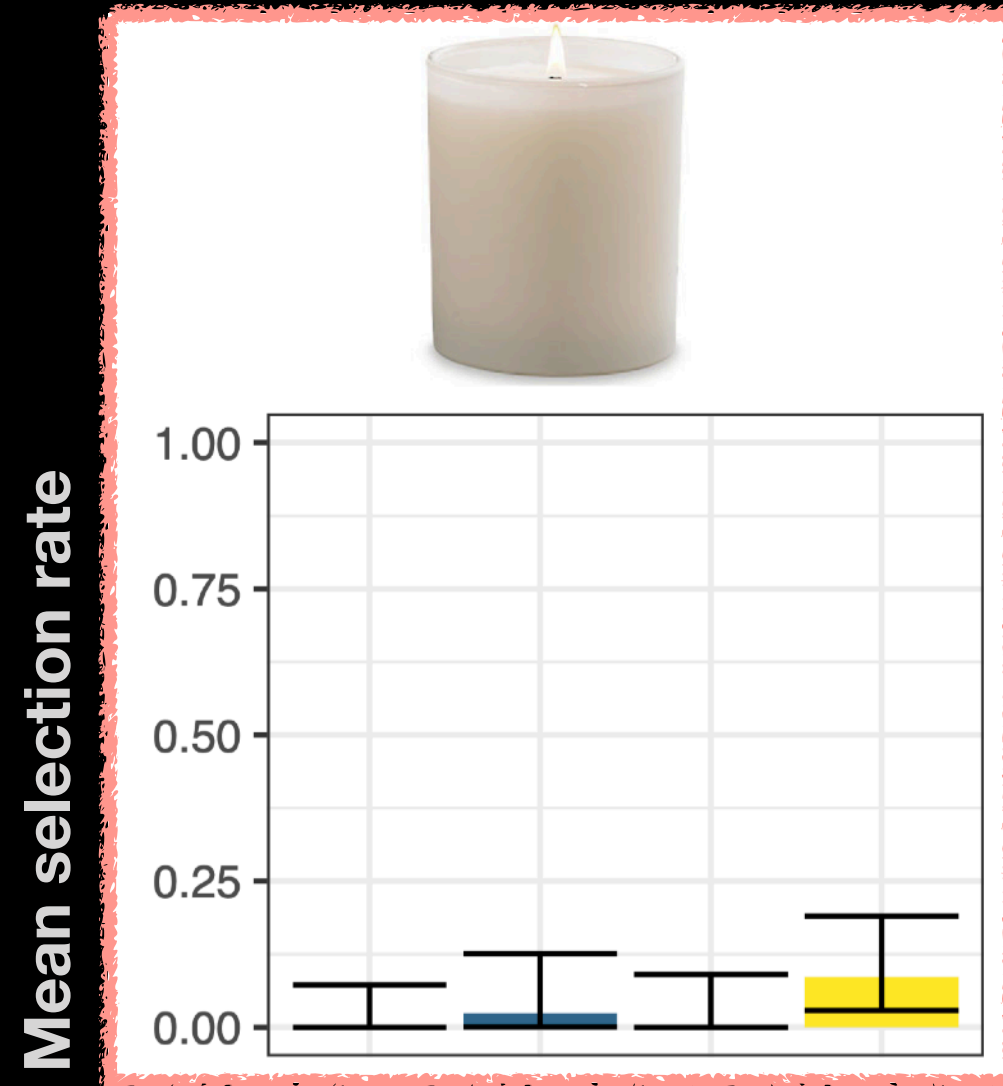
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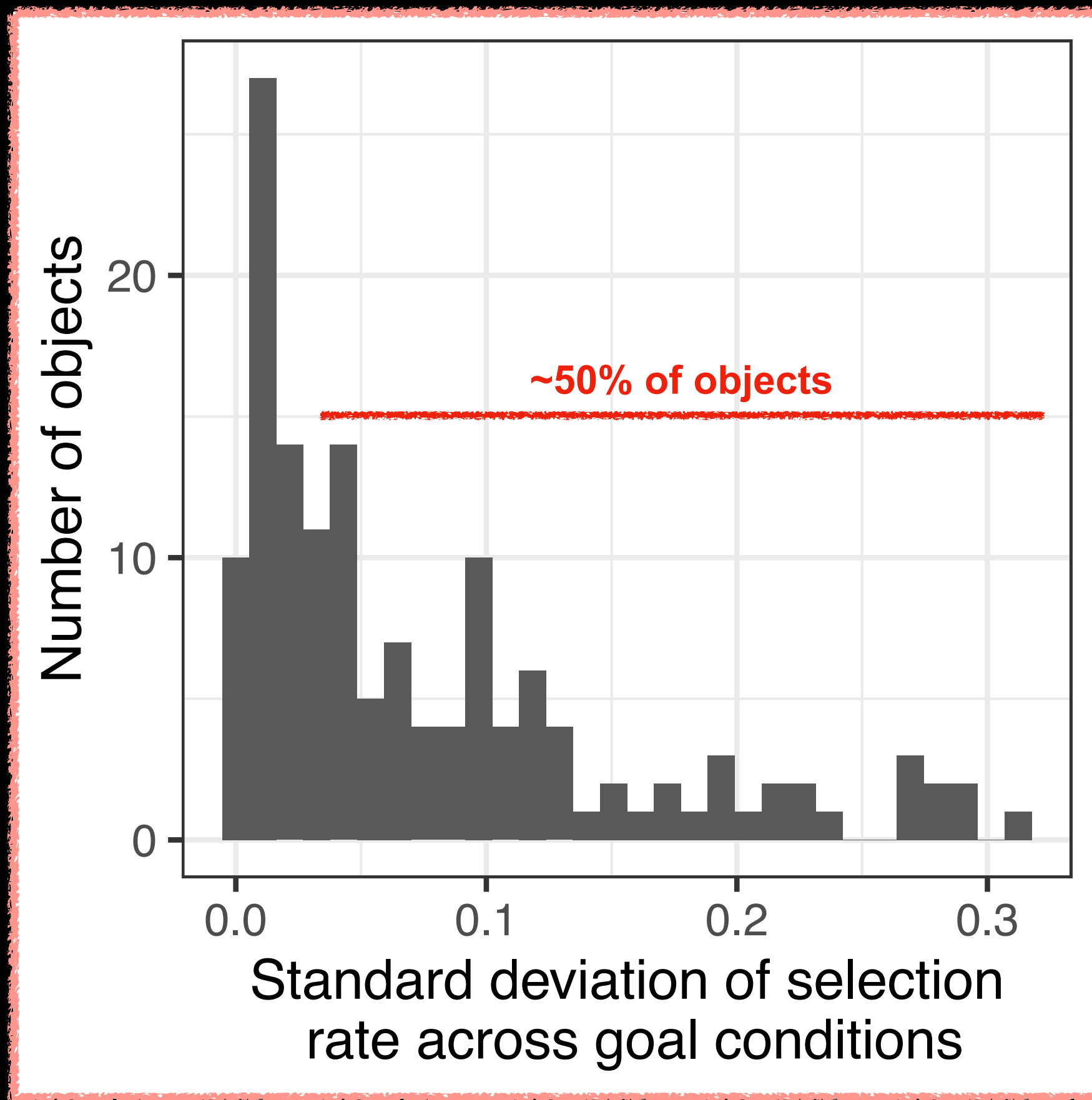
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- Formal characterizations of AN context dependence
- Experimental data to facilitate theorizing about ANs
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**Interpreters jointly infer intended meanings and non-communicative speaker goals under uncertainty**

(e.g. Kao et al. 2014a for **hyperbole**; Kao et al. 2014b for **metaphor**; Burnett 2019 for **social indexical meaning**)





(interpreter)

*rule*

**No electronic devices  
in the theater.**



(interpreter)

*rule*  
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**S**  
(signal producer)



**L<sub>1</sub>**  
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(interpreter)

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in the theater.

*obj*



*goal*

Limit distracting light

S

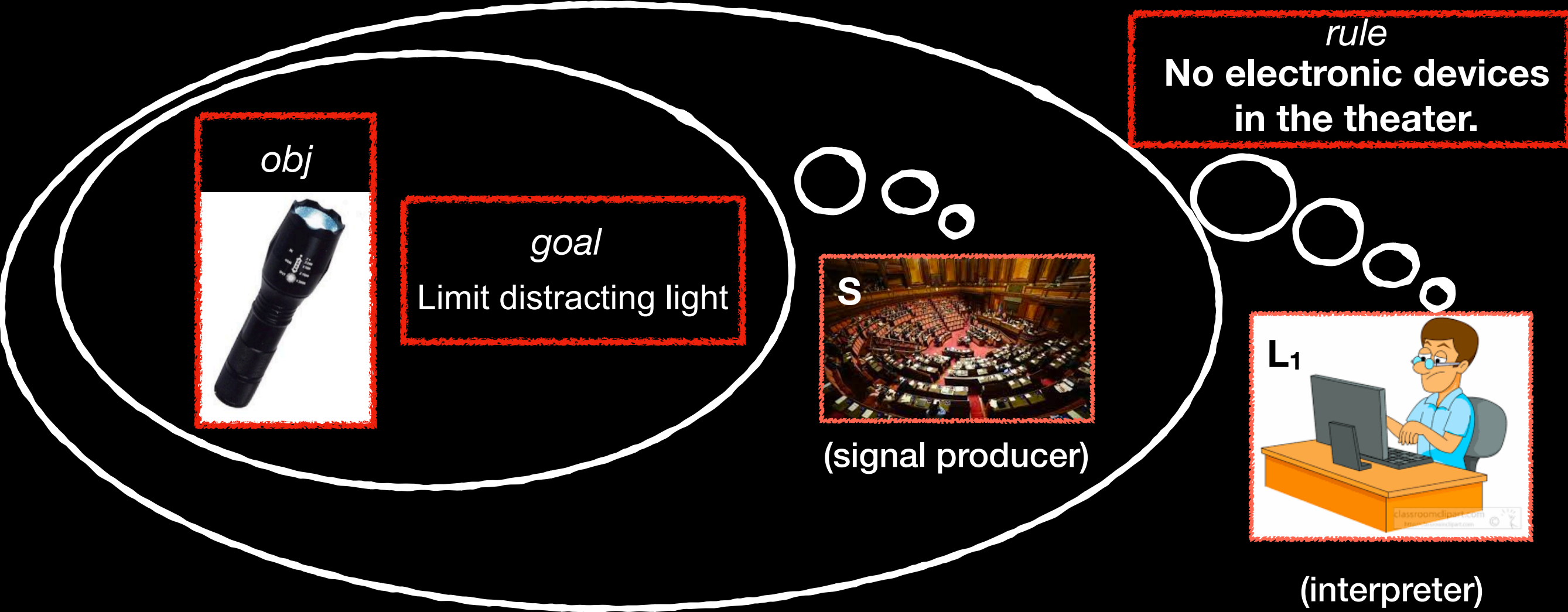


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L<sub>1</sub>



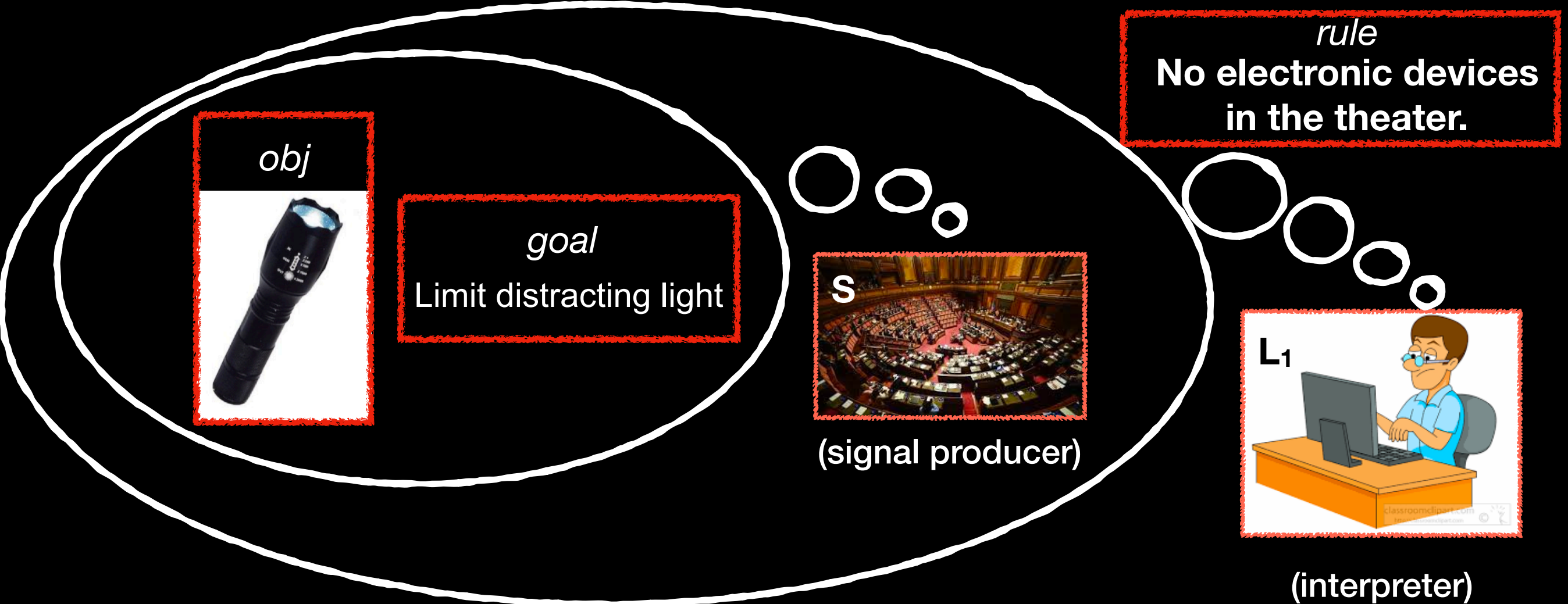
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$$L_1(obj, goal | rule)$$

Joint posterior probability that object *obj* is prohibited and signaler has policy *goal*, given observation of a *rule*.



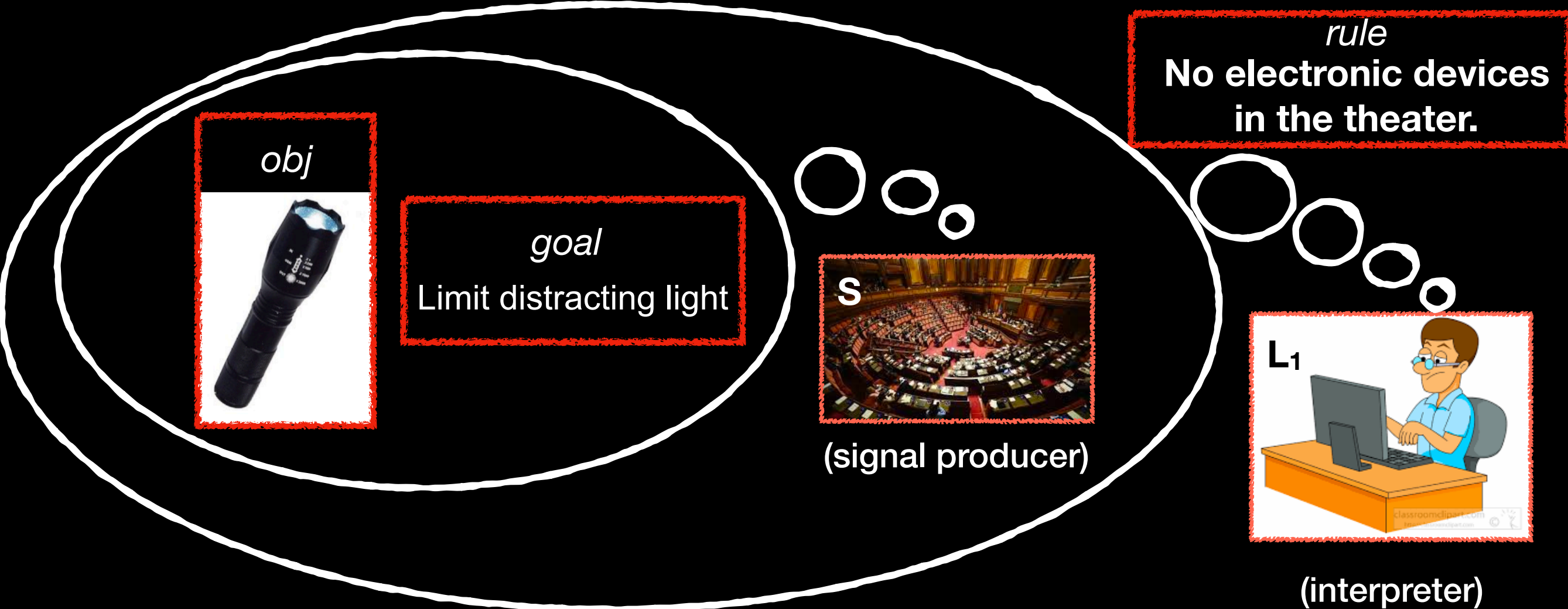


$$L_1(obj, goal | rule) \propto S(rule | obj, goal)$$

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Signaler probability of producing *rule*, given *goal* and intention of prohibiting *obj*.



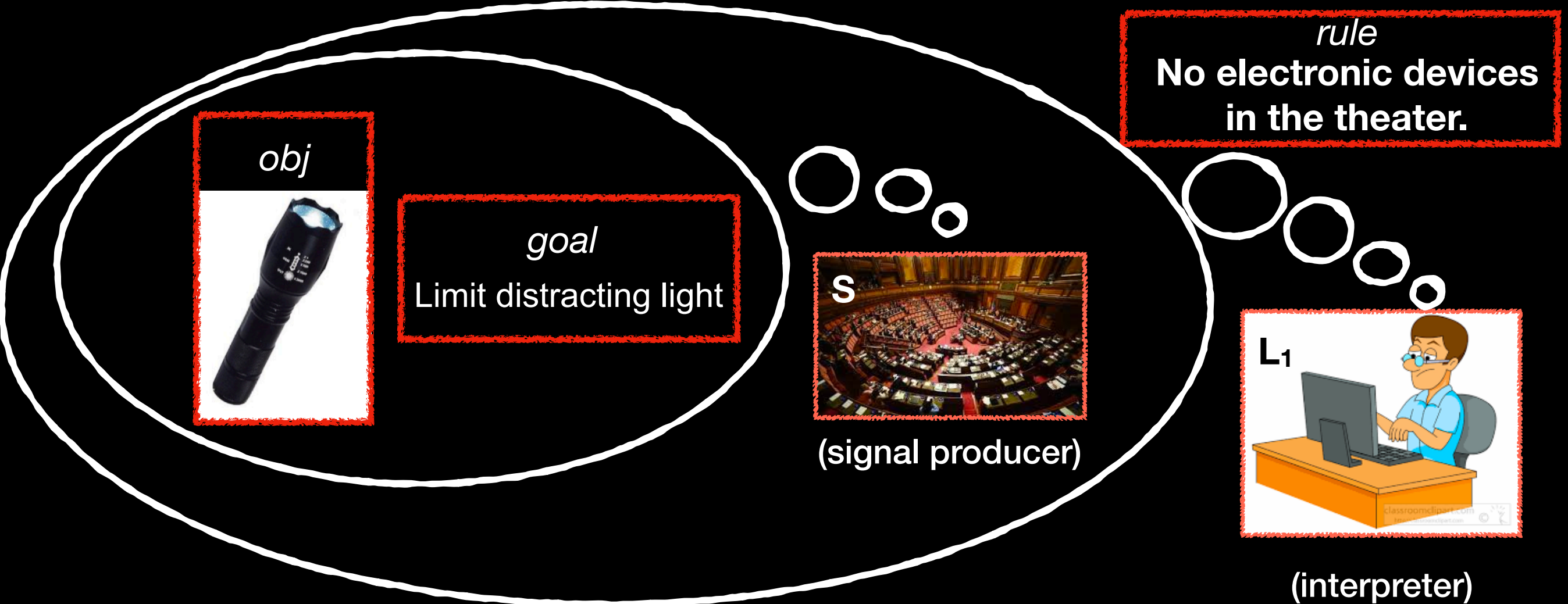


$$L_1(obj, goal | rule) \propto S(rule | obj, goal) \cdot P_G(goal | rule)$$

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$$L_1(obj, goal | rule) \propto S(rule | obj, goal) \cdot P_G(goal | rule) \cdot P_{CAT}(obj | rule)$$

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Prior beliefs that *obj* is prohibited, given lexical content of *rule*.

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Is this object an electronic device?

Definitely not.

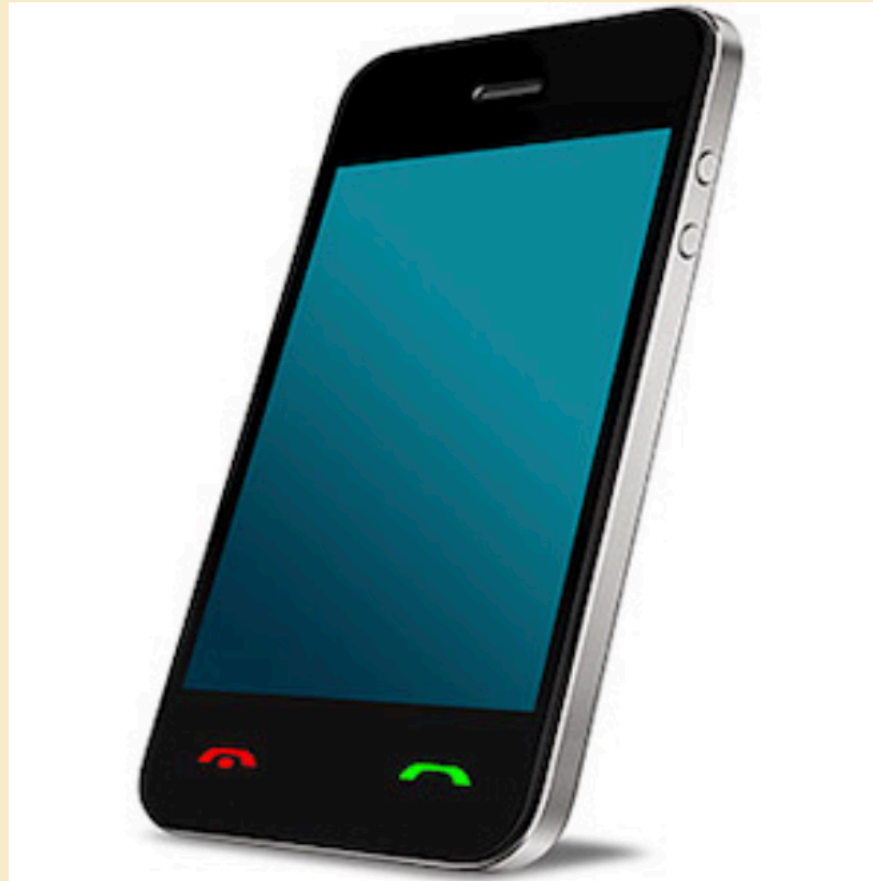
Definitely yes.

‘Category prior’ norming study (n = 100)



$$L_1(obj, goal | rule) \propto S(rule | obj, goal) \cdot P_G(goal | rule) \cdot P_{CAT}(obj | rule)$$

Signaler probability of producing *rule*, given *goal* and intention of prohibiting *obj*.



Does this object exhibit the following quality?  
**Could be used to record live performances.**

Definitely not.

Definitely yes.

‘Feature attribution’ norming study (n = 100)

$$L_1(obj, goal | rule) \propto S(rule | obj, goal) \cdot P_G(goal | rule) \cdot P_{CAT}(obj | rule)$$

$$\propto e^{\alpha \cdot \ln(U(goal, obj) - cost(rule))}$$

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- ... where  $U$  is bounded on  $[0,1]$  and tracks utility of prohibiting  $obj$  given  $goal$
- For  $obj$  on right:  
 $U(\text{pirate-recordings}, obj) \sim 0$   
 $U(\text{emit-light}, obj) \sim 1$



Our model: the Goal-Sensitive (GS) interpreter

$$L_1(obj, goal \mid rule) \propto S(rule \mid obj, goal) \cdot P_G(goal \mid rule) \cdot P_{CAT}(obj \mid rule)$$

Baseline: the Goal-Insensitive (GI) interpreter

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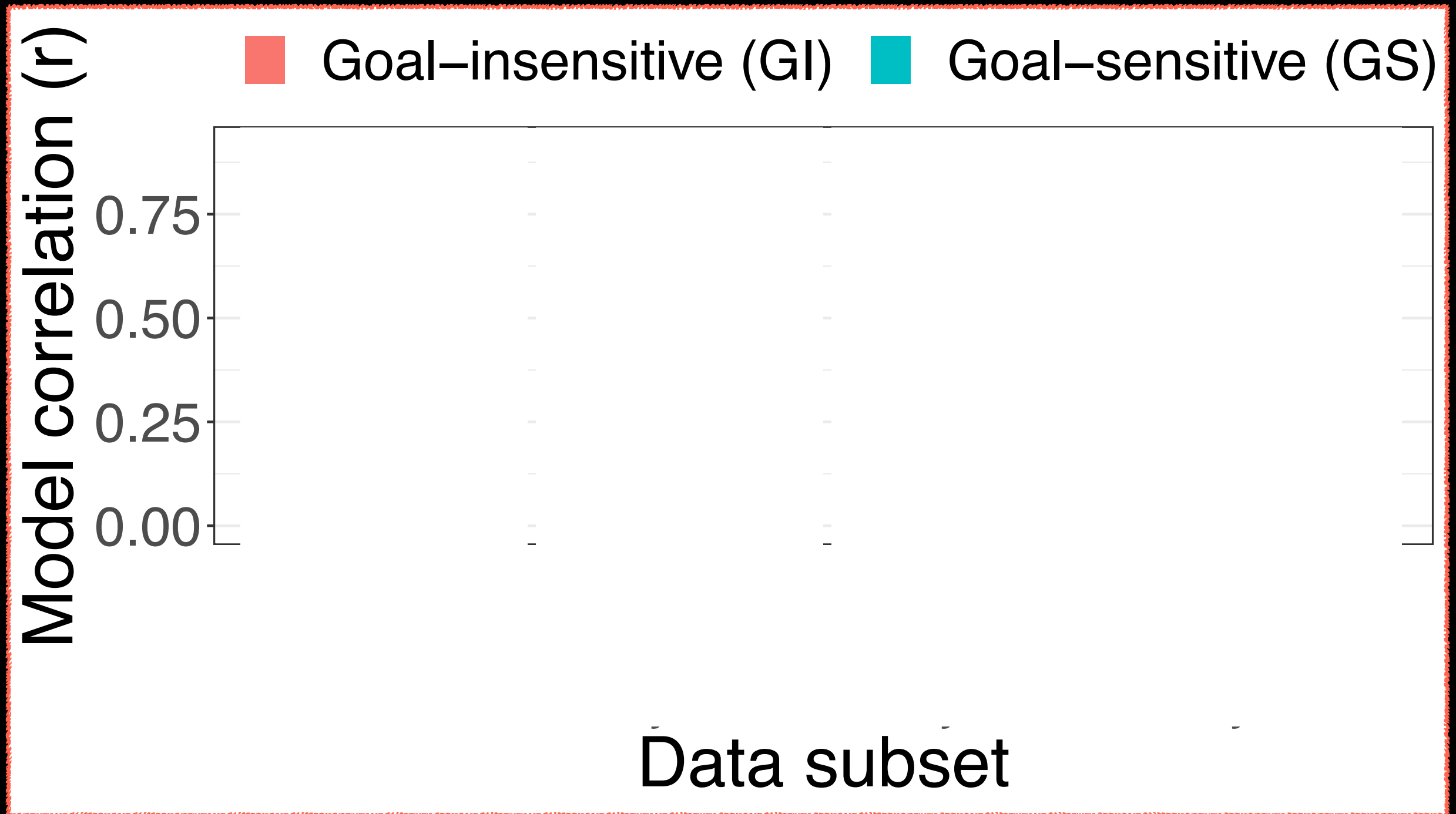
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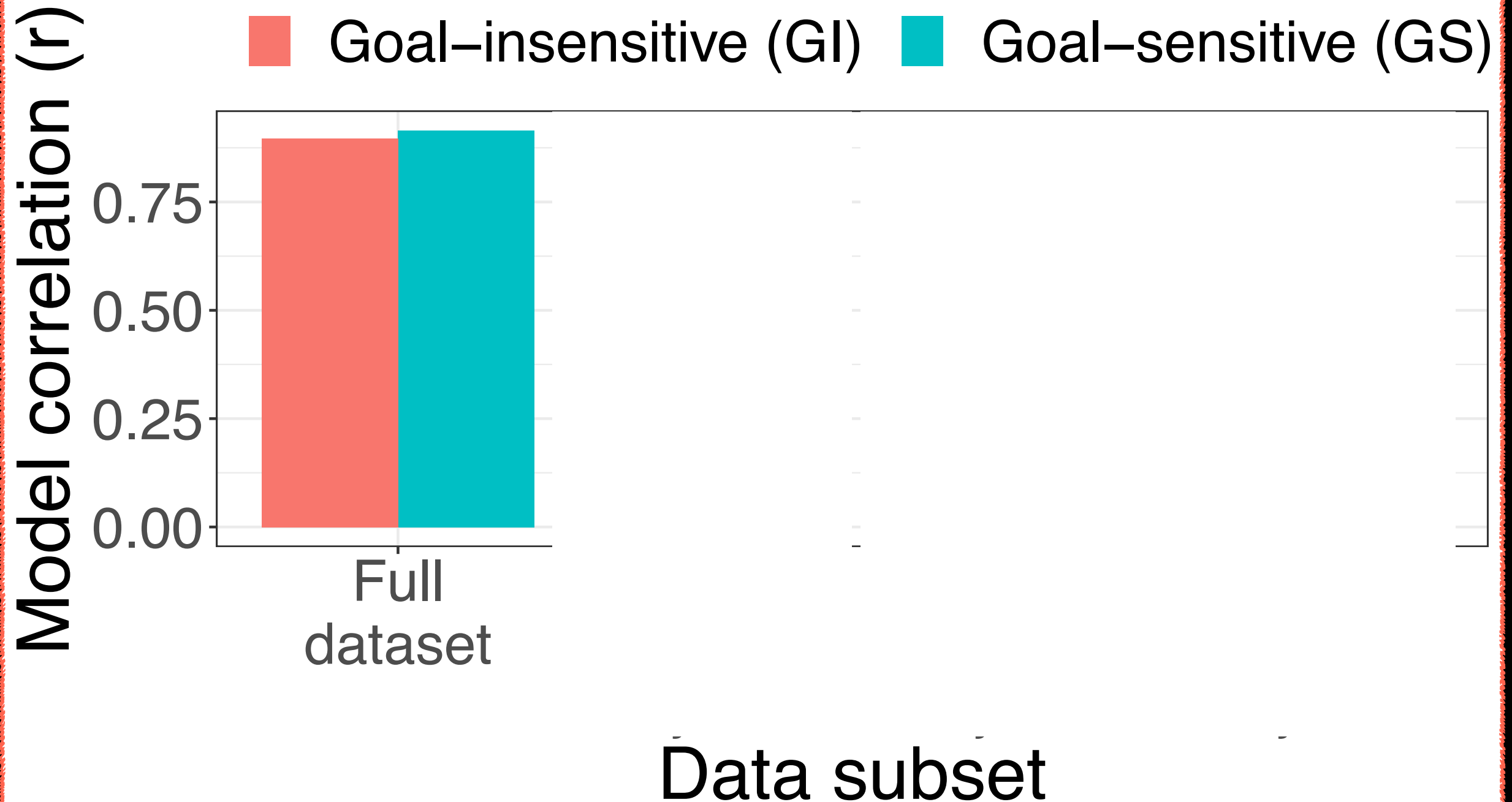
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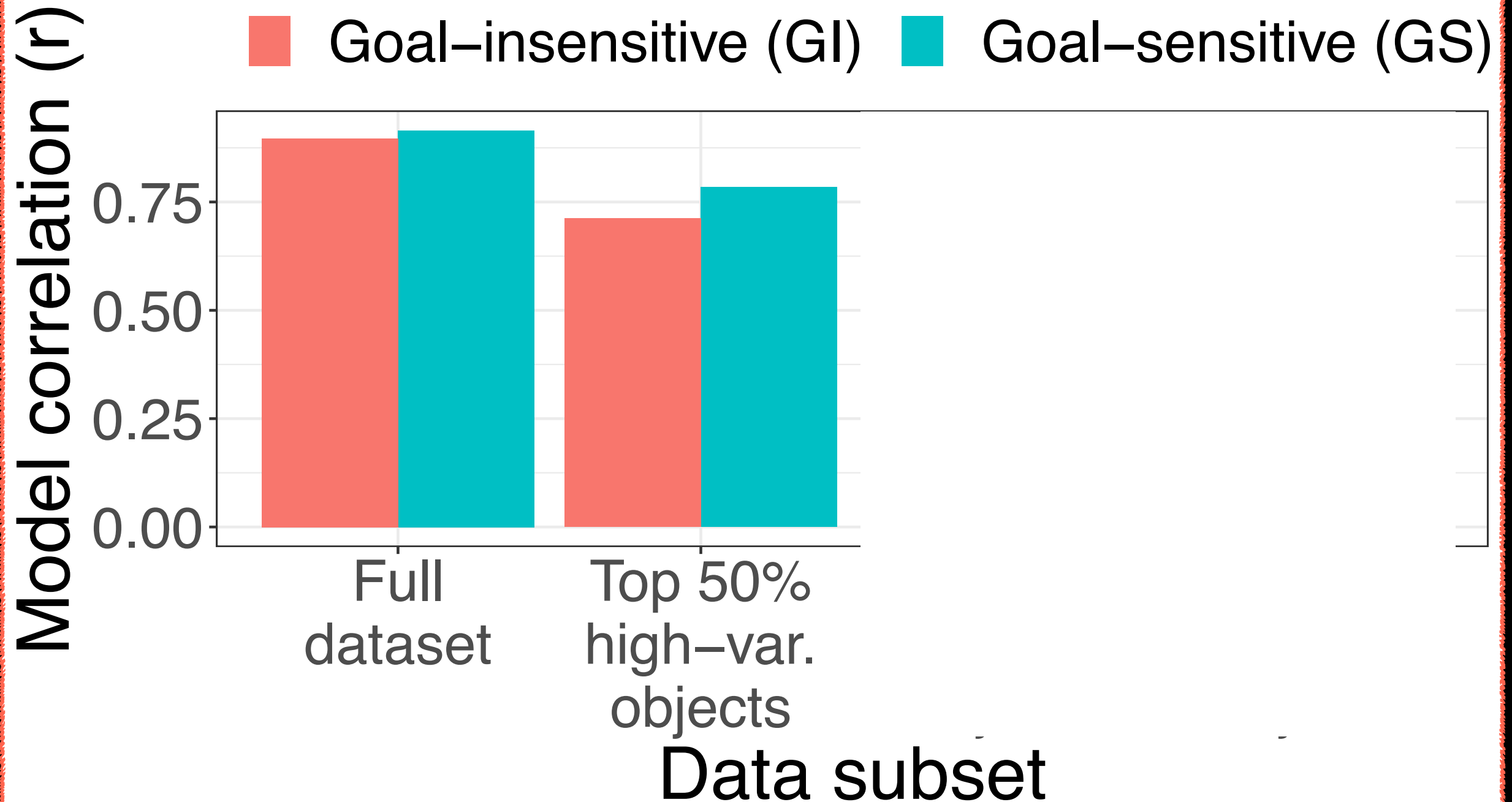
**Linking function:**

Proportion of *obj* selection ~ Marginal  $L_1$  prob. of *obj* prohibition

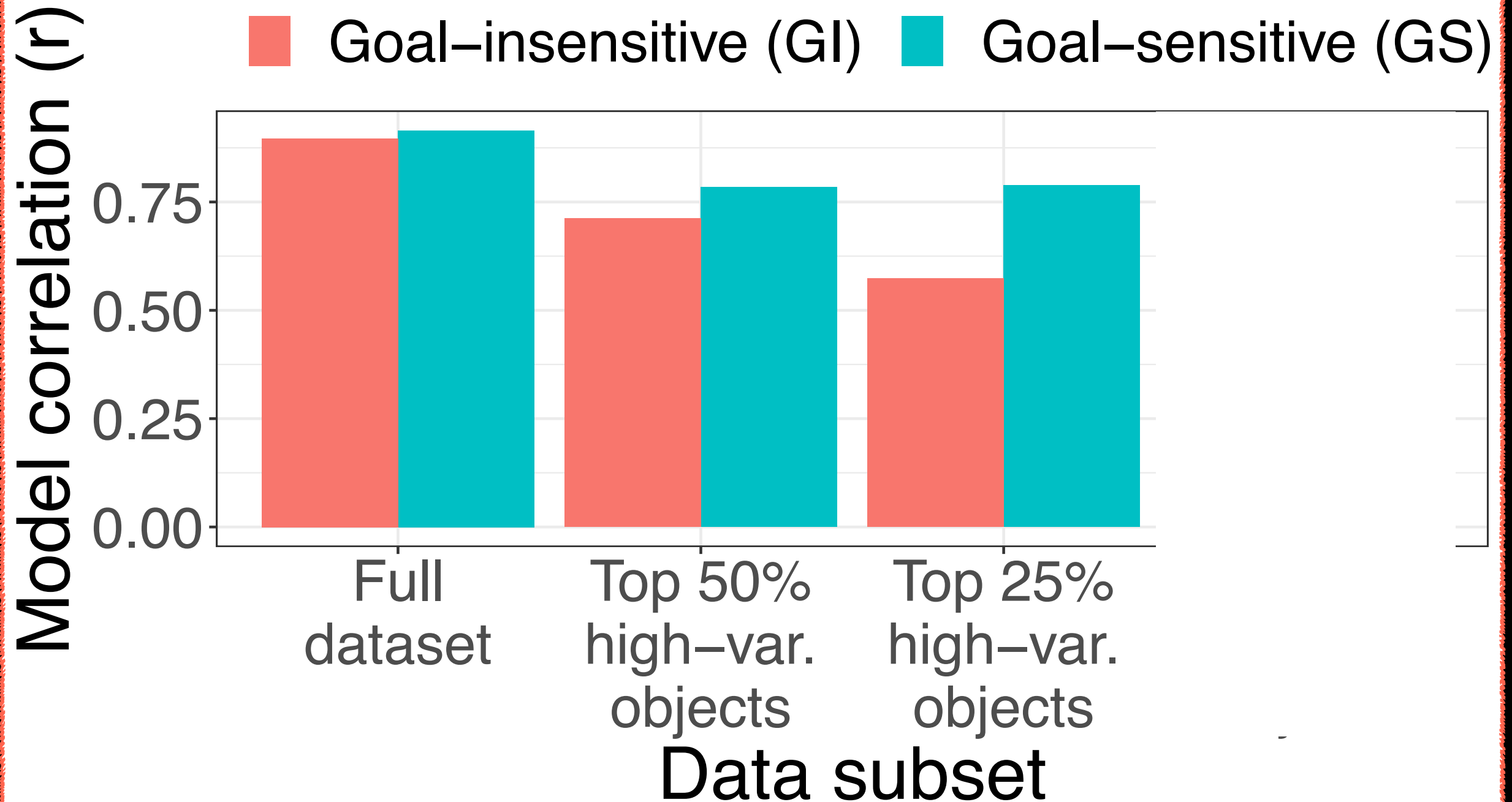




Goal-Sensitive model also favored via Bayesian model comparison  
(Bayes' Factor > 30)

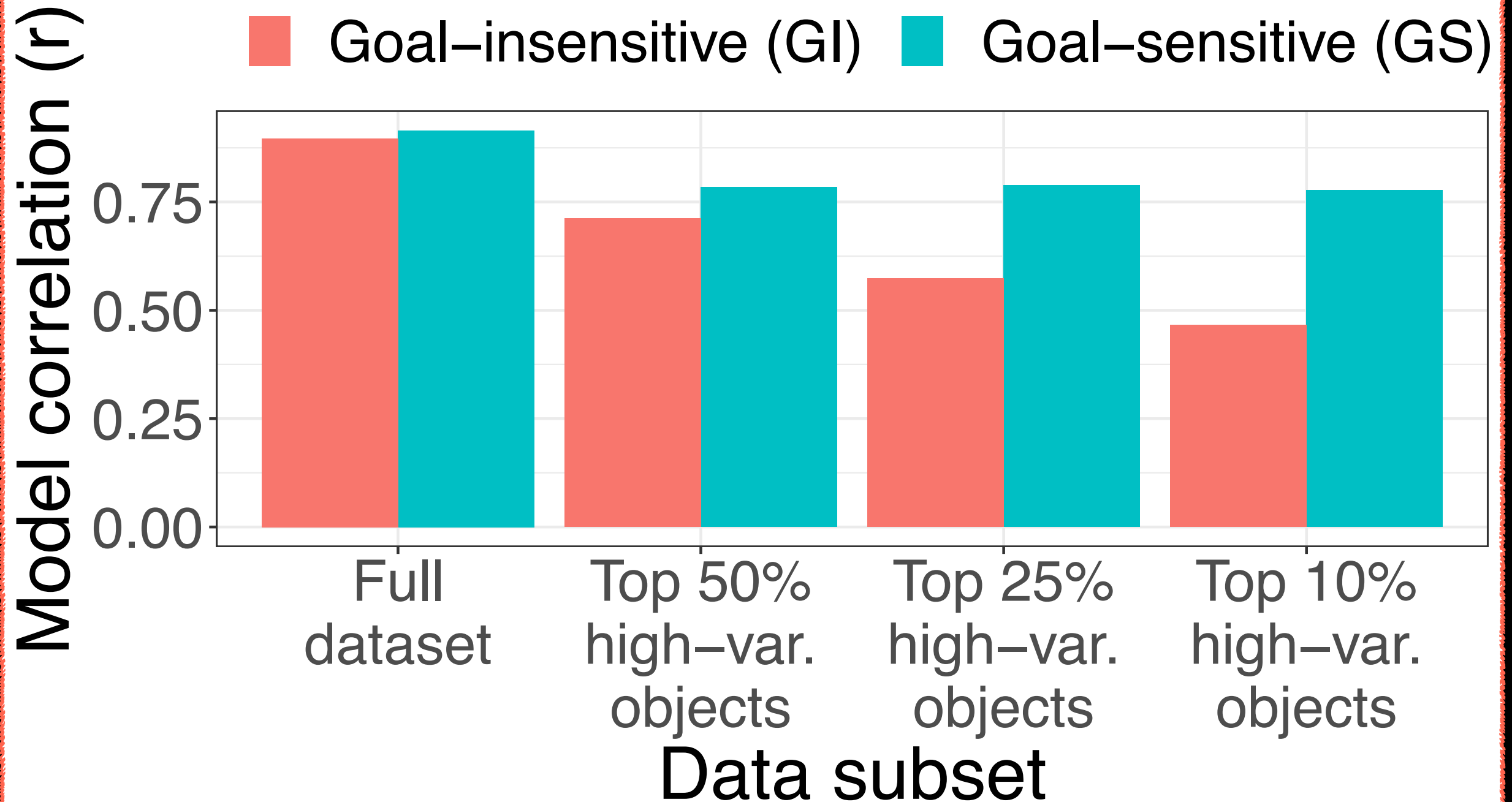


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## Model comparison



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  - ... modulation channel likely not (just) imprecision or domain restriction**
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**Semantic denotation of ANs requires specifying free semantic parameter(s) whose value(s) determine  $\langle e, t \rangle$ -type function in context (see e.g. Levin & Grimm, 2018)**

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# Conclusions

Proof of concept:

**How to extend inquiry into vague interpretation beyond single-dimensional expressions?**

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# Thanks for your attention!

bwaldon@stanford.edu  
@BRwaldon on Twitter



# Model details

$$L_1(obj, goal|rule)^1 \propto S(rule|obj, goal)^2 \cdot P_G(goal|rule)^3 \cdot P_{CAT}(obj|rule)^4$$

<sup>1</sup> $L_1(obj, goal|rule)$ : Joint posterior probability that object  $obj$  is prohibited and signaler has particular policy  $goal$ , given observation of a  $rule$ .

(Other possible prohibition state:  $\neg obj$  [ $obj$  is not prohibited])



# Model details

$^2S(rule|obj, goal)$ : Signaler prob. of producing *rule*, given *goal* and intention to prohibit *obj*.

$$\propto e^{\alpha \cdot \ln(U(goal, obj) - cost(rule))}$$

... where  $\alpha$  is an optimality parameter, *cost* is a cost function on signal choices, and  $U(goal, obj)$ :

- Tracks utility of prohibiting *obj* given a policy *goal*.
- Outputs on the interval [0,1] (the greater the extent to which prohibiting *obj* advances *goal*, the higher the output value).

$$U(goal, \neg obj) = 1 - U(goal, obj)$$

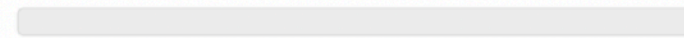
(Failing to prohibit objects that would advance *goal* is low utility).

- Is parameterized via a norming study in which objects in the experiment are normed for goal-relevant features, e.g.:

Does this object exhibit the following quality?  
**Could be used to record live performances.**



Definitely not.



Definitely yes.



# Model details

${}^3P_G(goal|rule)$ : Prior over policy goals (given observation of *rule*). Parameterized via a norming study in which policy goals are normed for a priori plausibility, e.g.:

**The managers of a theater are concerned that audience members might try to record performances and distribute pirate recordings online.**

How plausible is it that the **motivation** above could have given rise to the rule below?

No electronic devices are allowed in the theater.

Highly implausible.  Highly plausible.

${}^4P_{CAT}(obj|rule)$ : Prior beliefs that *obj* is prohibited, given lexical content of *rule*. (In particular, prior beliefs about the membership of *obj* in the category denoted by an artifact noun that features in the rule).

$$P_{CAT}(\neg obj|rule) = 1 - P_{CAT}(obj|rule)$$

Is this object an electronic device?



Definitely not.  Definitely yes.

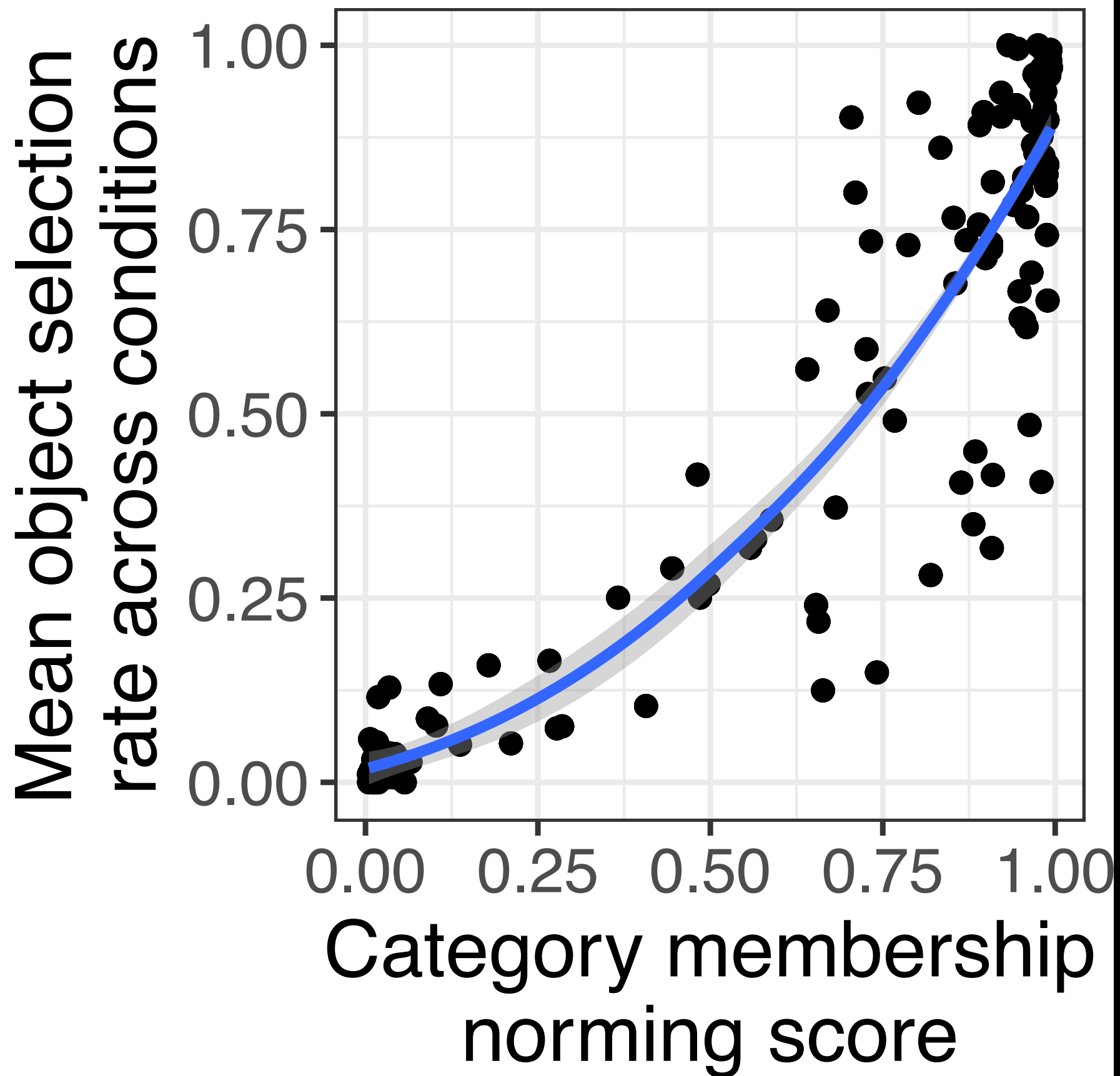


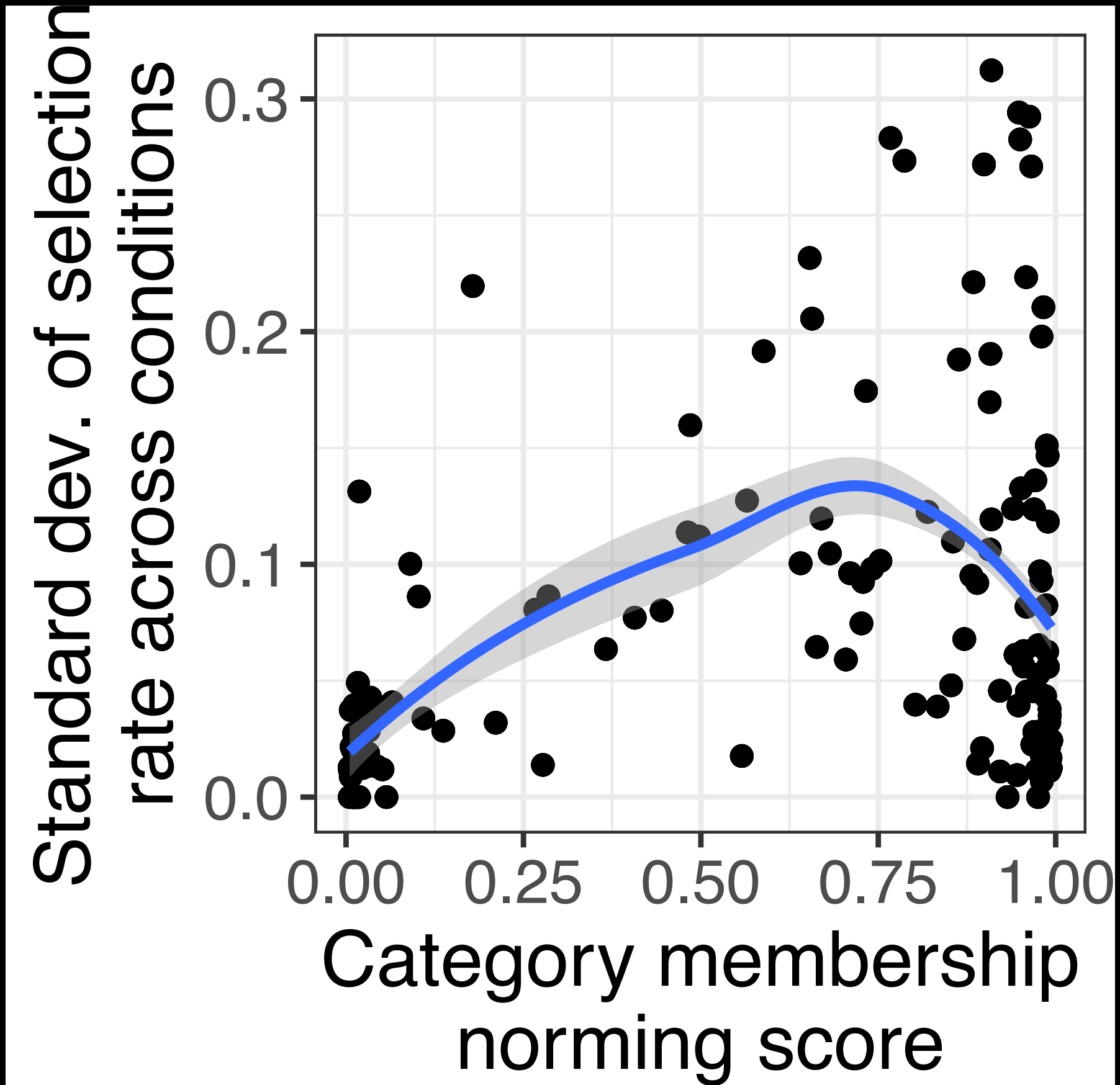
# Model comparison details

For the purposes of model comparison, we assume:

- A space of possible messages that includes a scene's featured *rule* and an alternative, *silence* (following Lassiter & Goodman, 2013) with the following properties:
  - $S(\text{silence}|\text{obj}, \text{goal}) = S(\text{silence}|\neg\text{obj}, \text{goal})$  for any *obj* and any *goal*.
  - For any  $\text{goal}_1, \text{goal}_2$  in the space of possible policy goals,  $P_G(\text{goal}_1|\text{silence}) = P_G(\text{goal}_2|\text{silence})$
  - $P_{CAT}(\text{obj}|\text{silence}) = P_{CAT}(\neg\text{obj}|\text{silence})$  for any *obj*
- For trials in which participants are exposed to an explicit policy goal  $\text{goal}_x$ ,  $P_G(\text{goal}_x|\text{rule}) = 1$ .
- $\text{cost}(\text{rule}) = \text{cost}(\text{silence}) = 0$

Model comparison was conducted in WebPPL (Goodman & Stuhlmüller, 2014) from a uniform prior over model architectures (goal-sensitive vs. goal-insensitive) and a uniform prior over  $\alpha$  values.





## Challenge for DR accounts:

- Experimental contexts permit explicit meta-linguistic commentary as to what 'counts as' a member of the category denoted by the noun as in (1), similar to negotiating gradable adjective thresholds (2) but unlike canonical DR contexts (3):

(1) *No electronic devices are allowed in the theater. (By the way: for our purposes, a flashlight [counts / doesn't count] as an electronic device.)*

(2) *Get me a long ladder. (By the way: for our purposes, 20 feet counts as long for a ladder.)*

(3) *(Planning a party): Put all the bottles of Heineken in the fridge. (# By the way: for our purposes, nothing that my neighbors bought for their party counts as a bottle of Heineken.)*

## Challenge for imprecision accounts:

- Some objects normed as category members but exhibit relatively low selection rates in some policy goal conditions.
- Ex: boombox in the *electronic devices* scene
- Account must posit that these objects are outside category but merely 'close enough' when relevant for advancing policy goals.